PROJECT PROPOSAL TEMPLATE

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Abstract

(200 words maximum) Follow the instructions, paying close attention to the word count limits and requirements. The abstract should state the motivation, that is, why the study will be done or which problem will be addressed, which method will be used, as well as the expected results and contributions or implications of the work. Give details. See for example the abstract in (Krizhevsky, Sutskever, & Hinton, 2017). In addition, include a graphical abstract as seen in Figure 1.

Abstract

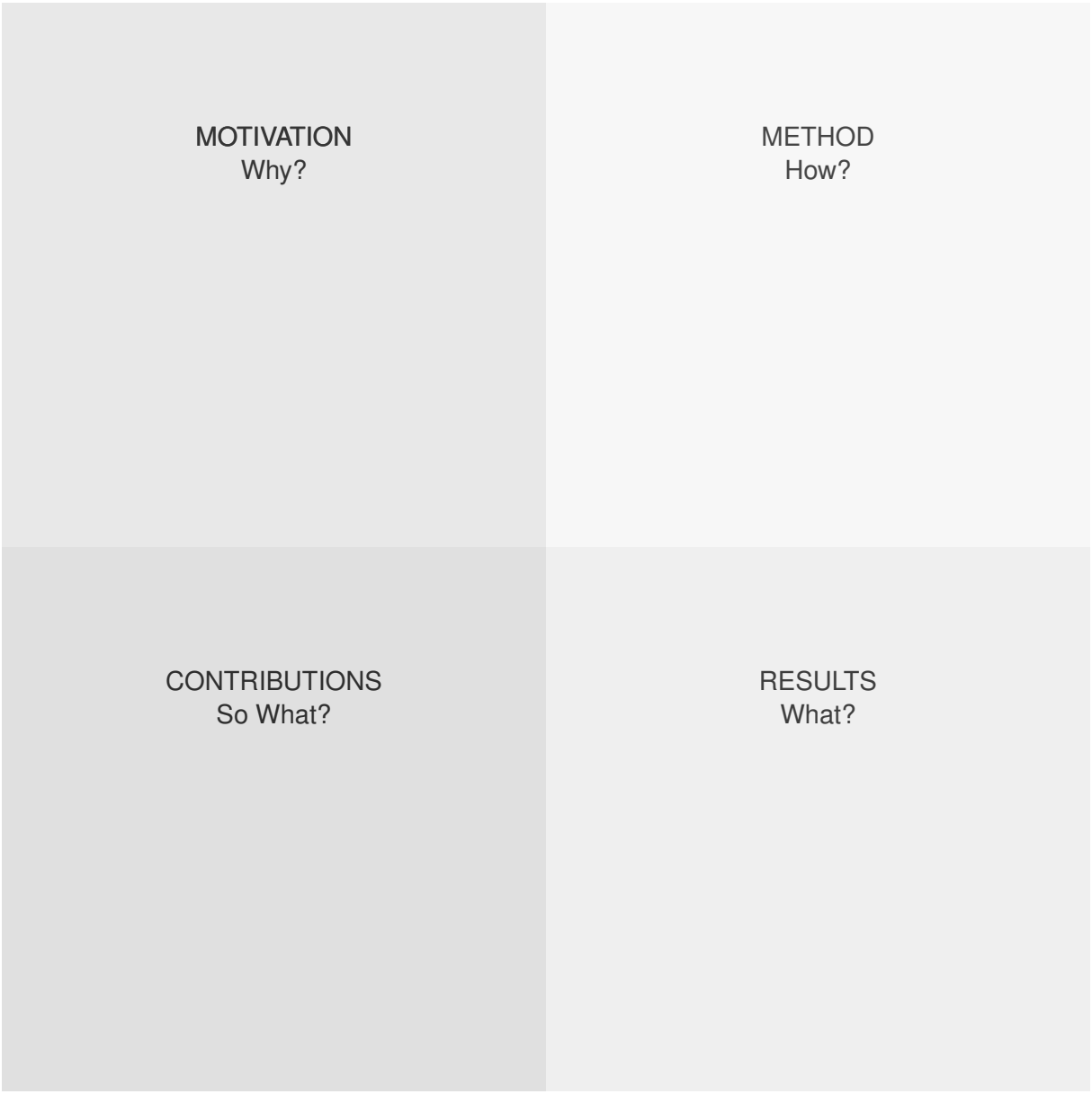


Figure 1: Graphical Abstract. Include a graphical abstract of the proposal with the motivation, method, expected results and contributions. Examples of graphical abstracts can be found in (Elsevier, 2024; Velarde et al., 2024).

1 Introduction

(30 words maximum.) Introduce the topic in one sentence. As the project advances, this template should be updated until it becomes the final document. Use the American Psychological Association (APA) style for citations (Hughes et al., 2017).

2 The State-of-the-art

(2 research papers minimum, 1 Comparison Table)

Present the State-of-the-art in a comparison table. Fill or adapt Table 1. For example, if the task is classification, the performance column should include the reported F1, Precision, and Recall on a given dataset or datasets. Add a comment if necessary.

Reference	Task	Performance	Dataset	Comment

Table 1: State-of-the-art comparison Table.

3 Objectives

(About 20 words per sentence.) Describe the main objective and sub-objectives of the project. Alternatively propose a hypothesis and its research questions.

- First sub-objective.
- Second sub-objective.
- Third sub-objective.

4 The method

(Include a Figure. Describe it in maximum 30 words.) Present the intended method graphically. Fill Figure 2.

4.1 Data

(Minimum 1 Dataset) List and describe the dataset or datasets to be used. Include the statistics: number of samples, features, etc.

Figure 2: Present the intended Method. Include the intended dataset, techniques, and evaluation framework. Add as much detail as possible in the figure. See for example Figure 1 in Sossi-Rojas et al. (2023, p. 2), or Figure 2 in Badrinarayanan et al. (2017, p. 4).

5 Results

(Include a table.) Present a comparison table with expected own results as in Table 2. See for example Tables 1 and 2 in Krizhevsky et al. (2017, p. 7). Indicate the task, metric and dataset. Provide the reference to the state-of-the-art results.

Model	**Top-1**	**Top-5**	**Dataset**	**Comment**
CNN (Krizhevsky et al., 2017)	37.5 %	17.0 %	ILSVRC-2010	On GTX 580 3GB GPUs
Expected own results	to be tested	to be tested	ILSVRC-2010	To be trained on Google Colab

Table 2: Expected results. Top-1 and Top-5 classification error rates obtained by a Convolutional Neural network (CNN) (Krizhevsky et al., 2017, p. 7) over ILSVRC-2010 test set (Berg et al., 2010).

6 Expected Contributions / Conclusion

(Maximum 5 sentences of less than 25 words each) List 2 to 5 expected contributions of your proposed project. This section can be renamed as ”Conclusion”.

- First expected contribution.
- Second expected contribution.
- Third expected contribution.

A The plan

(Include a Gantt chart with tasks and their expected duration) Include the specific dates of begin and end.

B Agreements and constraints

(Only if needed) State if the work needs a confidentiality agreement and mention the stakeholders if any.

References

Badrinarayanan, V., Kendall, A., & Cipolla, R. (2017). Segnet: A deep convolutional encoder-decoder architecture for image segmentation. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, *39*(12), 2481-2495. doi: 10.1109/TPAMI.2016.2644615

Berg, A., Deng, J., & Fei-Fei, L. (2010). *Large scale visual recognition challenge 2010.* (`https://www.image-net.org/`. Accessed: 27-10-2026.)

Elsevier. (2024). *Graphical abstract.* `https://www.elsevier.com/researcher/author/tools-and-resources/graphical-abstract`. ([Online; accessed 4-April-2024])

Hughes, J. L., Brannan, D., Cannon, B., Camden, A. A., & Anthenien, A. M. (2017). Conquering apa style: Advice from apa style experts. *Psi Chi Journal of Psychological Research*, *22*(3), 154–162.

Krizhevsky, A., Sutskever, I., & Hinton, G. E. (2017). Imagenet classification with deep convolutional neural networks. *Communications of the ACM*, *60*(6), 84–90.

Sossi-Rojas, S., Velarde, G., & Zieba, D. (2023). A machine learning approach for bitcoin forecasting. *Engineering Proceedings*, *39*(1). Retrieved from `https://www.mdpi.com/2673-4591/39/1/27` doi: 10.3390/engproc2023039027

Velarde, G., Weichert, M., Deshmunkh, A., Deshmane, S., Sudhir, A., Sharma, K., & Joshi, V. (2024). Tree boosting methods for balanced and imbalanced classification and their robustness over time in risk assessment. *Intelligent Systems with Applications*, *22*, 200354. Retrieved from `https://www.sciencedirect.com/science/article/pii/S2667305324000309` doi: https://doi.org/10.1016/j.iswa.2024.200354